Gursimran Singh

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Research Summary

Ph.D. researcher in **programmable network systems and security**. I design and evaluate line-rate security functions (deep packet inspection, firewalls, intrusion detection) in **P4 on Intel Tofino** switches, characterize the hardware limits that constrain them, and build tools that measure **Internet censorship and resilience**. Two first-author publications; committee service for EuroSys, PPoPP, ICS, CoNEXT, and ACM S3; adjunct teaching; industry work on intent-based networking.

Interests: programmable data planes (P4, Tofino, SmartNICs/FPGAs) · in-network security and collective computation · Internet measurement, censorship, and resilience under disaster · security and privacy · dependable systems under hardware limits.

Education

Oklahoma State University — **Jan 2026 – Present**
Ph.D. in Computer Science · Advisor: Dr. H. B. Acharya — *Expected May 2028*

Rochester Institute of Technology — **2021 – 2025**
Ph.D. coursework, Computing and Information Sciences · GPA 3.80/4.0 · transferred to OSU with advisor — *Rochester, NY*

University at Buffalo (SUNY) — **2019 – 2021**
M.S. in Computer Science and Engineering · GPA 3.52/4.0 — *Buffalo, NY*

University of Delhi, Sri Guru Gobind Singh College of Commerce — **2016 – 2019**
B.Sc. (Hons.) Computer Science — *New Delhi, India*

Publications

[P1] **Gursimran Singh**, H. B. Acharya, and Minseok Kwon. **Programmable Data Planes for Network Security.** In *Proc. 17th International Conference on Networks & Communications* (*NeTCoM 2025*), Computer Science & Information Technology (CS & IT) vol. 15, no. 13, pp. 163–180, July 2025. DOI: 10.5121/csit.2025.151314 · arXiv:2507.22165.
Systematizes security on P4 switches across attack mitigation, firewalls and middleboxes, in-network cryptography, and in-switch machine learning; identifies the workarounds (recirculate-and-truncate, lookup-table precomputation) that make these functions feasible under memory and instruction-set limits, and the open problems that remain.

[P2] **Gursimran Singh** and H. B. Acharya. **POSTER: A Cyberspace Study of the Russia–Ukraine War.** In *Proc. ACM ASIA Conference on Computer and Communications Security* (*ASIA CCS 2023*), pp. 1016–1018, July 2023. DOI: 10.1145/3579856.3592822.
Measured hard and soft failures of website access from Ukraine, Russia, and the United States (March–July 2022) using a new soft-failure classifier and a multi-protocol traceroute; found a substantial change in access to sites from Ukraine over the period, with no significant collateral damage from filtering by Russian ASes.

Additional first-author manuscripts are currently under submission (details available upon request).

Research Experience

Doctoral Researcher, advised by Dr. H. B. Acharya — **2021 – Present**
Oklahoma State University (2026 – present) · Rochester Institute of Technology (2021 – 2025)

Programmable data-plane security on Intel Tofino — 2021 – Present

- Design and implement application-layer security functions (deep packet inspection and pattern matching, stateful filtering, NIDS-style detection) in P4 on Intel Tofino and Tofino2 switches (Tofino Native Architecture), engineering around per-stage memory, restricted instruction sets, and parser limits.
- Develop packet-analysis and filtering algorithms that use recirculation, cross-packet flow state, and precomputed lookups to approach firewall and intrusion-detection capability at line rate; prototype and evaluate on Tofino hardware (P4 Studio SDE, BF-RT control plane).
- First-authored a systematization of programmable-data-plane security research (NeTCoM 2025, [P1]).
- *Active directions:* correctness of inspection under recirculation and flow-state pressure (inspected versus merely forwarded throughput); safe evolution of security state across heterogeneous data paths (ASIC, SmartNIC/DPU, FPGA, host) with exactly-once enforcement and recovery; minimum equivalent firewall policies with checkable optimality certificates; orchestration and failure handling for in-network computing.

Internet censorship and resilience measurement — 2022 – Present

- Designed and implemented measurement tooling for the Russia–Ukraine war study: a classifier that detects *soft* failures of website access (beyond outright connection failures) and a multi-protocol traceroute implementation that localizes where interference occurs along a path.
- Collected and analyzed longitudinal measurements from Ukraine, Russia, and the United States (March–July 2022), attributing reachability changes to network conditions versus filtering (ACM ASIA CCS 2023, [P2]).
- *Active directions:* sustainability of refraction-networking deployments when operators withdraw, recovering access without losing resistance to blocking; probe-induced bias in censorship and resilience measurement.

Industry Experience

Keysight Technologies **Sep 2026 – Present**

R&D Software Engineering Intern (FPGA development for AI infrastructure testing solutions) · part-time, remote

- Software-side contributor to an intent-based networking effort that translates natural-language intent and protocol specifications into C/C++ implementations targeted for FPGA realization, working alongside FPGA-focused interns.
- Designed the intent-to-implementation architecture: intent/specification ingestion, a C/C++ intermediate path, and a proposed direct intent-to-HDL path.

Teaching Experience

Adjunct Instructor, Rochester Institute of Technology **2022, 2024 – 2025**

CSEC-140 Intro to Cybersecurity (2022, 2025) · CSCI-455 Principles of Cybersecurity (2024, 2025) · GCIS-123 Software Dev. & Problem Solving I (2025)

- Instructor of record: delivered lectures and hands-on virtual labs covering authentication, data security, risk management, network, system, and application security, and human factors.
- Taught computational problem solving in Python: searching and sorting, file I/O, lists, classes, and dictionaries.

Teaching Assistant, Rochester Institute of Technology **2023 – 2024**

CSCI-243 Mechanics of Programming · CSEC-759 Advanced Networking & Distributed Systems Security (graduate seminar)

Teaching Assistant, Oklahoma State University **Spring 2026**

Distributed and Cloud Computing

Research & Capstone Mentor **2022 – 2025**

Mentored three master's students and two undergraduate students on research and capstone projects

Professional Service

Artifact Evaluation Committee, EuroSys 2027 (both submission cycles) 2026 – 2027
Artifact Evaluation Committee, ACM PPoPP 2027 2026 – 2027
Junior Program Committee, ACM ICS 2027 (International Conference on Supercomputing) 2026 – 2027
Technical Program Committee, ACM S3 2026 student workshop (co-located with ACM MobiCom 2026) 2026
Shadow Technical Program Committee, ACM CoNEXT 2026 2026

Leadership

President, Computer Science Graduate Student Association (CS-GSA), Oklahoma State University: lead a seven-officer board; orientation, professional-development talks, networking, travel/funding advocacy 2026 – 2027
Council Member, Doctoral Student Association, Rochester Institute of Technology 2025
Vice-President, CSE Graduate Student Association (CSEGSA), University at Buffalo 2020 – 2021
President (2018 – 2019) and **Vice-President** (2017 – 2018), ACE (Computer Science Society), SGGSCC, University of Delhi; organizer, CYTHON-18 tech fest 2017 – 2019

Competitions & Honors

MITRE Embedded CTF (eCTF) 2021, University at Buffalo team: implemented the team's RSA (512-bit) key-generation and primality-testing routines in C for a secure embedded wireless link 2021
3rd place, 2020 regional round (hosted by RIT's ESL Global Cybersecurity Institute), Collegiate Penetration Testing Competition (CPTC), University at Buffalo team Nov 2020
"Stars of the College" feature, SGGSCC Annual Report, for academics and student leadership 2018

Technical Skills

Languages: C, C++, $P4_{16}$, Python, Java, Bash; CUDA and MATLAB (project experience)

Programmable networking: Intel Tofino / Tofino2 (TNA), P4 Studio SDE, BF-RT control plane, parsing/deparsing, recirculation, mirroring, match-action pipeline design; exposure to BMv2, eBPF/XDP, SmartNIC/DPU and FPGA data paths

Measurement & security tooling: Scapy, Wireshark, Zmap, Nmap, traceroute-based path measurement, Aircrack-ng, Kali Linux, IDA, OllyDbg

Systems & data: Linux, Git, VirtualBox lab networks, Pintos kernel development; scikit-learn, Keras, NumPy, Pandas

Certifications & Training

Intel Connectivity Academy: ICA-XFG-101 ($P4_{16}$, Tofino family, P4 Studio SDE); ICA-XFG-201 – 206 (Advanced TNA: parsing/deparsing, recirculation, mirroring) 2023 – 2024
OffSec Wireless Professional (OSWP) and **Kali Linux Certified Professional (KLCP)**, Offensive Security 2023
Google Cloud: Introduction to Generative AI, Responsible AI, and Large Language Models 2023

Selected Projects

Reliable transport protocols and multi-client TCP chat (C/C++): Alternating-Bit, Go-Back-N, and Selective-Repeat over an unreliable channel.

Task offloading for QoS with a dynamic ANN (C, C++, CUDA): run-time adaptation to query parameters (max hops, query volume).

Core OS components in Pintos (C): kernel-level threading and the system-call interface.